

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

CHEMISTRY — SSC-I (9th Grade) — ANSWER KEY (Version 1)

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

1. Chemistry deals with the composition, structure, properties, and changes of:

Answer: (C) Matter 1 mark

Chemistry is defined as the science that investigates the materials of the universe and the changes that these materials undergo. It deals with the **composition, structure, properties, behavior, and changes of matter** and energy. Understanding the fundamental concepts of chemistry helps to explain natural phenomena.

2. The branch of chemistry that deals with carbon-containing substances (except carbonates, bicarbonates, oxides, carbides) is:

Answer: (C) Organic Chemistry 1 mark

Organic chemistry is the branch of chemistry that deals with substances containing carbon (except carbonates, bicarbonates, oxides, and carbides). Inorganic chemistry deals with elements and their compounds except organic compounds.

3. The branch that focuses on the study of polymers, their types, properties, and uses is:

Answer: (B) Polymer Chemistry 1 mark

Polymer chemistry is the branch of chemistry that focuses on the study of polymers, their types, properties, uses, importance, and types of polymerizations. Examples of synthetic polymers include nylon bearings, plastic bags, polyethylene cups, polyester, Teflon-coated cookware, and epoxy glue.

4. The integration of scientific knowledge for human needs is known as:

Answer: (C) Technology 1 mark

Technology is the process of applying scientific knowledge to practical applications, resulting in the creation of tools, machines, and systems that enhance our lives. The integration of scientific knowledge for human needs is known as technology. Science is the systematic study of nature, while engineering is the application of scientific principles to construct systems.

5. Forensic chemistry is the application of:

Answer: (B) Analytical Chemistry 1 mark

Forensic chemistry is the application of **analytical chemistry**. It involves the examination of physical traces such as body fluids, bones, fibers and drugs. It can be used to identify an unknown compound. Analytical chemistry deals with methods and instruments for determining the composition and properties of matter.

6. Anything that has mass and occupies space is called:

Answer: (D) Matter 1 mark

Anything that has **mass** and occupies **space** is called **matter**. This makes air, water, rocks, and even people examples of matter. Matter is itself composed of the atom, which is the building block of all matter.

7. A highly ionized gas where electrons are separated from atoms is called:

Answer: (D) Plasma 1 mark

Plasma is a highly ionized gas where electrons are separated from atoms, often found in high-energy environments. Plasmas are electrically conductive. Examples include the sun, lightning, welding arc, solar wind, etc.

8. Bose-Einstein Condensate (BEC) occurs at temperatures near:

Answer: (C) -273.14°C 1 mark

Bose-Einstein Condensate (BEC) is a state of matter that occurs at near **absolute zero temperatures** (-273.14°C), where particles occupy the same quantum state, acting as a single quantum entity. For example, BEC of rubidium atoms at extremely low temperatures.

9. Diamond, graphite, and buckyballs are allotropes of:

Answer: (C) Carbon 1 mark

Diamond, graphite, and buckyballs (fullerenes/C-60) are three important **allotropes of carbon**. Allotropy is the property of an element to exist in different physical forms. These different forms in the same physical state are called allotropes.

10. A solution containing maximum amount of solute at a given temperature is:

Answer: (D) Saturated 1 mark

A **saturated solution** is a solution containing the **maximum amount of solute** at a given temperature. At this stage, the solvent cannot dissolve any more solute. Dynamic equilibrium is established between dissolution and crystallization at this point.

11. The scattering of light by colloidal particles is called:

Answer: (C) Tyndall effect 1 mark

The **Tyndall effect** is the scattering of a beam of light by colloidal particles. We can see the path of scattered light beam inside the colloidal solution. This is the main characteristic which distinguishes colloids from true solutions. Colloids are also called false solutions or colloidal solutions.

12. Water is called a universal solvent because it:

Answer: (B) Dissolves most compounds 1 mark

Water is called a **universal solvent** because it **dissolves the majority of compounds** present in earth's crust. Aqueous solutions are mostly used in laboratories. Aqueous solution is formed by dissolving a substance in water.

SECTION B — Short Questions (11 × 3 = 33 Marks)

Q.2(i): Define chemistry. List any four branches of chemistry with their definitions. 3 marks

Chemistry is defined as the science that investigates the materials of the universe and the changes that these materials undergo. It deals with the composition, structure, properties, behavior, and changes of matter and energy. 1 mark

Four branches of chemistry: ½ mark each = 2 marks

1. **Organic Chemistry:** Deals with substances containing carbon (except carbonates, bicarbonates, oxides, and carbides).
2. **Inorganic Chemistry:** Deals with elements and their compounds except organic compounds.
3. **Physical Chemistry:** Deals with laws and theories to understand the structure and changes of matter.
4. **Analytical Chemistry:** Deals with methods and instruments for determining the composition and properties of matter.

OR

Q.2(i) OR: What is green chemistry? Why is it important in reducing pollution? 3 marks

Green chemistry is the model of chemical products and processes that reduce the use of **hazardous substances**. 1 mark

Importance: 2 marks

- Green chemistry is helpful in understanding and reducing **pollution**.
- It aims to design products and processes that minimize the generation of hazardous waste.
- It promotes the use of safer chemicals and renewable resources, reducing the environmental footprint of chemical processes.

Q.2(ii): Differentiate between organic chemistry and inorganic chemistry. 3 marks

Property	Organic Chemistry	Inorganic Chemistry
Definition	Deals with substances containing carbon (except carbonates, bicarbonates, oxides, and carbides) 1 mark	Deals with elements and their compounds except organic compounds 1 mark
Examples	Methane (CH ₄), Ethanol (C ₂ H ₅ OH)	Sodium chloride (NaCl), Iron oxide (Fe ₂ O ₃)
Application	Organic chemists synthesize new medicines that interact with proteins or enzymes 1 mark	Lithium-ion (Li-ion) batteries are an application of inorganic chemistry

OR

Q.2(ii) OR: Differentiate between geochemistry and astrochemistry. 3 marks

Property	Geochemistry	Astrochemistry
Definition	Deals with the study of chemical composition, distribution, and transformation of elements and compounds in the Earth's crust 1½ marks	Deals with the study of chemical processes and reactions that occur in astronomical environments such as stars, planets, comets, and interstellar space 1½ marks
Examples	Rocks, minerals, soils, water, atmosphere	Chemical reactions in stars and comets

Q.2(iii): Differentiate between science and technology with examples. 3 marks

Property	Science	Technology
Definition	Science is the systematic process of constructing and organizing knowledge about the universe. It seeks to understand the natural world. 1 mark	Technology is the process of applying scientific knowledge to practical applications, resulting in the creation of tools, machines, and systems. 1 mark
Purpose	To explore and understand natural phenomena	To apply knowledge for human needs
Example	Scientists study the principles of photovoltaic cells to understand how sunlight is converted into electricity 1 mark	Technologists develop solar panels based on the scientific principles discovered

OR

Q.2(iii) OR: Define engineering. How do chemical engineers contribute to industry? 3 marks

Engineering is the use of science and mathematics to design and construct systems, structures, and tools for various processes. 1 mark

Contribution of chemical engineers: 2 marks

- Chemical engineers develop and design **manufacturing processes** for the production of chemicals, fuels, food, medicines, polymers, detergents, and paper.
- They work to maximize **productivity** and **product quality**.
- They ensure chemical products are produced efficiently and safely for industrial and consumer use.

Q.2(iv): Explain with an example how science, technology, and engineering work together to solve real-world problems. 3 marks

Example: Harnessing Solar Energy

- **Science:** Scientists study the principles of photovoltaic cells to understand how sunlight can be converted into electricity. 1 mark
- **Technology:** Technologists develop solar panels and energy storage systems based on the scientific principles discovered. 1 mark
- **Engineering:** Engineers play their role in designing and implementing large-scale renewable energy systems. For instance, an electrical engineer might design the wiring and connections of a solar power plant, and a civil engineer could design the infrastructure. 1 mark

Together, science, technology, and engineering contribute to the development and utilization of sustainable energy sources.

OR

Q.2(iv) OR: Describe two daily life applications of chemistry from different branches. 3 marks

1. **Organic Chemistry — Medicine:** To treat diseases, organic chemists synthesize new medicines that interact with specific targets like proteins or enzymes. 1½ marks
2. **Environmental Chemistry — Water Protection:** Environmental chemistry is used to protect water that has been poisoned by soil and dust by using different methods e.g., sedimentation, filtration, and disinfection. 1½ marks

Q.2(v): Define matter. List the four commonly known states of matter with one example each. 3 marks

Matter: Anything that has **mass** and occupies **space** is called matter. 1 mark

Four states of matter: ½ mark each = 2 marks

1. **Solid:** Has definite shape and volume. Particles tightly packed. E.g., ice, diamond, metals.
2. **Liquid:** Has definite volume but takes shape of container. Particles close but can move. E.g., water, milk, oil.
3. **Gas:** Has neither definite shape nor volume. Particles far apart. E.g., oxygen (O₂), hydrogen (H₂).
4. **Plasma:** Highly ionized gas, electrically conductive. E.g., the sun, lightning, welding arc.

OR

Q.2(v) OR: What is Bose-Einstein Condensate (BEC)? What are liquid crystals? Give one example of each. 3 marks

Bose-Einstein Condensate (BEC): A state of matter that occurs at near **absolute zero temperatures** (-273.14°C), where particles occupy the same quantum state, acting as a single quantum entity. For example, BEC of **rubidium atoms** at extremely low temperatures. 1½ marks

Liquid Crystals: Intermediate states between liquids and solids, where molecules flow like a liquid but retain some ordered structure. They are commonly used in display technology. For example, **Liquid Crystal Displays (LCDs)**. 1½ marks

Q.2(vi): Compare the macroscopic properties (density, compressibility, fluidity) of solids, liquids, and gases. 3 marks

Physical Property	Solid	Liquid	Gas
Density	High density at normal condition	High density at normal condition	Low density at normal condition due to large spaces between molecules 1 mark
Compressibility	Not compressible	Moderately compressible	Very compressible because of large empty spaces 1 mark
Fluidity	Cannot flow	Can flow	Can flow 1 mark

OR

Q.2(vi) OR: Define plasma. Give three examples of plasma found in nature or daily life. 3 marks

Plasma: Plasma is a highly ionized gas where **electrons are separated from atoms**, often found in high-energy environments. Plasmas are **electrically conductive** and are affected by electrical and magnetic fields. 1½ marks

Three examples: ½ mark each = 1½ marks

1. The **Sun** (and other stars)
2. **Lightning**
3. **Welding arc** / Solar wind

Q.2(vii): Define element, compound, and mixture with one example each. 3 marks

Element: A pure substance made up of only **one type of atom**. It cannot be broken down into simpler substances by chemical means. Example: Oxygen (O₂), Gold (Au), Carbon (C). 1 mark

Compound: A pure substance made up of **two or more elements chemically combined** in a fixed ratio. Compounds have different properties from the elements that make them. Example: Water (H₂O), Sodium chloride (NaCl). 1 mark

Mixture: A combination of two or more substances (elements or compounds) that are **physically mixed together** but not chemically combined. They do not have a fixed ratio and can be separated by physical means. Example: Air (a mixture of nitrogen, oxygen, carbon dioxide), Saltwater. 1 mark

OR

Q.2(vii) OR: Differentiate between compound and mixture (any three differences). 3 marks

Property	Compound	Mixture
Combination	Elements are chemically combined in a fixed ratio 1 mark	Substances are physically mixed , not in a fixed ratio
Separation	Can only be separated into elements through chemical reactions 1 mark	Can be separated by physical means (filtration, evaporation, etc.)
Properties	Has different properties from its constituent elements 1 mark	Components retain their individual properties

Q.2(viii): What is allotropy? Name and briefly describe three allotropes of carbon. 3 marks

Allotropy: The property of an element to exist in **different physical forms** is called allotropy. These different forms in the same physical state are called **allotropes**. Atoms of the same element are arranged in different manners in allotropes. 1 mark

Three allotropes of carbon:

1. **Diamond:** The hardest and purest crystalline allotrope of carbon. Each carbon atom is bonded to four others forming a rigid tetrahedral network. Non-conductor of electricity. $\frac{3}{4}$ mark
2. **Graphite:** Composed of flat two-dimensional layers of hexagonally arranged carbon atoms. Weak intermolecular bonds between layers make it soft and slippery. Good conductor of electricity. $\frac{3}{4}$ mark
3. **Buckyballs (C-60):** Also known as fullerenes. Have a football-like fused hollow ring structure made up of twenty hexagons and twelve pentagons. Each of its 60 carbon atoms is bonded to 3 other carbon atoms. $\frac{1}{2}$ mark

OR

Q.2(viii) OR: Describe the structure and properties of diamond. Why is it non-conductive?

3 marks

Structure: Diamond is the hardest and the purest **crystalline allotrope of carbon**. In its structure, each carbon atom is **covalently bonded to four other carbon atoms** forming a rigid network of **tetrahedral shape**. This three-dimensional arrangement makes it the hardest substance with a very high melting point. $1\frac{1}{2}$ marks

Properties: Diamond is extremely hard, has a very high melting point, and is transparent. $\frac{1}{2}$ mark

Non-conductivity: Since all the carbon atoms are bonded with other carbon atoms, **no free electrons** are present in the structure. The absence of free electrons makes diamond a **non-conductor of electricity**. 1 mark

Q.2(ix): Define solution, solute, and solvent. Give one example of each. 3 marks

Solution: A solution is a **homogeneous mixture** of two or more substances in which one substance is dissolved in the other. Homogeneous means that no particles or parts of different substances can be seen. The particles in solution are microscopic, less than 1 nm in diameter. Example: salt dissolved in water.

1 mark

Solute: A substance that is **dissolved** is called a solute. Example: In salt solution, **salt** is the solute.

1 mark

Solvent: A substance in which the solute is **dissolved** is called a solvent. Example: In salt solution, **water** is the solvent. 1 mark

OR

Q.2(ix) OR: What is an aqueous solution? Why is water called a universal solvent? 3 marks

Aqueous Solution: An aqueous solution is formed by **dissolving a substance in water**. The dissolved substances in an aqueous solution may be solids, gases, or other liquids. In order to be a true solution, a mixture must be stable. For example, sugar in water and table salt in water. $1\frac{1}{2}$ marks

Universal Solvent: Water is called a **universal solvent** because it **dissolves the majority of compounds** present in the earth's crust. Aqueous solutions are mostly used in laboratories. $1\frac{1}{2}$ marks

Q.2(x): Differentiate between saturated and unsaturated solutions. 3 marks

Property	Saturated Solution	Unsaturated Solution
Definition	A solution containing maximum amount of solute at a given temperature 1 mark	A solution which contains lesser amount of solute than that which is required to saturate it at a given temperature 1 mark
Dissolving capacity	Cannot dissolve any more solute; further added solute remains undissolved	Has the capacity to dissolve more solute
Equilibrium	Dynamic equilibrium is established between dissolution and crystallization 1 mark	No equilibrium; more solute can be dissolved

OR**Q.2(x) OR: What is a supersaturated solution? How can it be prepared? 3 marks**

Supersaturated Solution: A solution that contains **greater amount of solute** than is required to form a saturated solution. It is more concentrated than a saturated solution. Supersaturated solutions are **not stable**. **1½ marks**

Preparation: An easy way to get a supersaturated solution is to prepare a **saturated solution at high temperature** (by heating and dissolving more solute). It is then **cooled to a lower temperature** where the excess solute crystallizes out and leaves behind a saturated solution. If you add a crystal of sodium thiosulphate to a supersaturated solution, crystallization will start. **1½ marks**

Q.2(xi): Differentiate between colloid and suspension (any three differences). 3 marks

Property	Colloid	Suspension
Particle size	1 to 1000 nm. Not visible by naked eye 1 mark	Greater than 1000 nm (10^3 nm). Visible by naked eye
Filtration	Particles can pass through normal filter paper but not through ultra-filter paper 1 mark	Particles cannot pass through normal as well as ultra-filter paper
Stability	Does not separate on standing 1 mark	Separates or settles down when stationary
Examples	Milk, starch, blood, soap solutions	Chalk in water, muddy water, paints

OR**Q.2(xi) OR: Explain the effect of temperature on the solubility of solutes with examples.****3 marks**

The solubility of solutes depends on temperature. Depending on the nature of solute, there is either:

1 mark

a) Increase in solubility with temperature: Most solid solutes show an increase in solubility when temperature increases. For example, the solubility of **KCl** increases from 34.7g (at 20°C) to 56g (at 100°C) per 100g of water. Similarly, **NH₄Cl** increases from 37.5g to 77g. **1 mark**

b) Decrease in solubility with temperature: Some solutes show a decrease in solubility with increasing temperature. For example, **Ca(OH)₂** decreases from 0.173g (at 20°C) to 0.066g (at 100°C) per 100g of water. Similarly, **Na₂SO₄** also decreases. Calcium hydroxide can be used to treat chemical burns and as an antacid. **1 mark**

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3(i): Describe any five branches of chemistry with their definitions and one daily life application of each. 5 marks

Chemistry is a diverse field of study with numerous sub-fields and interdisciplinary areas. Five important branches are:

1. Organic Chemistry: 1 mark

Definition: The branch of chemistry that deals with substances containing carbon (except carbonates, bicarbonates, oxides, and carbides).

Daily Life Application: To treat diseases, organic chemists synthesize new medicines that interact with specific targets like proteins or enzymes.

2. Inorganic Chemistry: 1 mark

Definition: The branch of chemistry that deals with elements and their compounds except organic compounds.

Daily Life Application: Lithium-ion (Li-ion) batteries are used as rechargeable batteries for electronics, toys, wireless headphones, handheld power tools, and electric vehicles.

3. Analytical Chemistry: 1 mark

Definition: The branch of chemistry that deals with the methods and instruments for determining the composition and properties of matter.

Daily Life Application: Forensic chemistry is the application of analytical chemistry. It involves examining physical traces such as body fluids, bones, fibers, and drugs to identify unknown compounds.

4. Environmental Chemistry: 1 mark

Definition: The branch of chemistry that deals with the study of chemical and toxic substances that pollute the environment and their adverse effects on human beings.

Daily Life Application: Environmental chemistry is used to protect water that has been poisoned by soil and dust by using different methods e.g., sedimentation, filtration, and disinfection.

5. Physical Chemistry: 1 mark

Definition: The branch of chemistry that deals with laws and theories to understand the structure and changes of matter.

Daily Life Application: The batteries in our vehicles are built on the principle of electrochemistry, which is a part of physical chemistry.

OR

Q.3(i) OR: Explain the relationship between science, technology, and engineering with at least two detailed examples from the field of chemistry. 5 marks

Definitions:

- **Science:** The systematic process of constructing and organizing knowledge about the universe. Science seeks to understand the natural world. **1 mark**
- **Technology:** The process of applying scientific knowledge to practical applications, resulting in the creation of tools, machines, and systems. The integration of scientific knowledge for human needs. **½ mark**
- **Engineering:** The use of science and mathematics to design and construct systems, structures, and tools for various processes. **½ mark**

Example 1: Harnessing Solar Energy 1½ marks

Scientists may study the principles of photovoltaic cells to understand how sunlight can be converted into electricity. Technologists can develop solar panels based on the scientific principles discovered. Engineers play their role in designing and implementing large-scale renewable energy systems. For instance, an electrical engineer might design the wiring and connections of a solar power plant, a civil engineer could be involved in designing the infrastructure. Together, science, technology, and engineering contribute to the development and utilization of sustainable energy sources.

Example 2: Organic Chemistry in Action — French Fries 1½ marks

The oil used to fry potatoes contains carbohydrates, which are organic molecules. Scientists study carbohydrates to learn more about how they work, so food technologists extract oil from seeds. Chemical engineers design oil production equipment and processes so that oil is produced efficiently and safely for

cooking. This shows how science provides the understanding, technology transforms it into practical applications, and engineering implements it at scale.

Q.3(ii): Describe the states of matter in detail. Include solid, liquid, gas, and plasma with their properties and examples. 5 marks

A state of matter refers to the distinct physical forms that matter can take, characterized by different physical properties like shape, volume, and the arrangement of particles. The most commonly known states of matter are:

1. Solid: 1 mark

A solid has a **definite shape and volume**. Particles are tightly packed in a **fixed arrangement**. Solids have high density at normal condition, are **not compressible**, and **cannot flow**. Examples: Ice, diamond, graphite, metals.

2. Liquid: 1 mark

A liquid has a **definite volume** but takes the **shape of its container**. Particles are close together but can move past one another. Liquids have high density, are **moderately compressible**, and **can flow**. Examples: Water, oils, milk, honey.

3. Gas: 1 mark

A gas has **neither a definite shape nor volume**. Particles are far apart and move freely. Gases have low density at normal condition due to large spaces between molecules, are **very compressible**, and **can flow**. Examples: Oxygen (O₂), Hydrogen (H₂), Nitrogen (N₂).

4. Plasma: 1 mark

Plasma is a **highly ionized gas** where electrons are separated from atoms, often found in high-energy environments. Plasmas are **electrically conductive** and are affected by electrical and magnetic fields. Examples: The sun, lightning, welding arc, solar wind.

Macroscopic Properties Table: 1 mark

Property	Gas	Liquid	Solid
Density	Low	High	High
Compressibility	Very compressible	Moderately compressible	Not compressible
Fluidity	Can flow	Can flow	Cannot flow

Note: States of matter can be converted from one to another through physical techniques such as changing the temperature or pressure. However, the chemical composition remains the same.

OR

Q.3(ii) OR: What are allotropes? Explain the three allotropes of carbon (diamond, graphite, and buckyballs) in detail including their structure and properties. 5 marks

Allotropy: The property of an element to exist in different physical forms is called **allotropy**. These different forms in the same physical state are called **allotropes**. Atoms of the same element are arranged in different manners in allotropes. They are different structural forms of the same element. **1 mark**

1. Diamond: 1½ marks

Structure: Diamond is the hardest and the purest crystalline allotrope of carbon. In its structure, each carbon atom is **covalently bonded to four other carbon atoms** forming a rigid network of **tetrahedral shape**. This three-dimensional arrangement makes it the hardest substance.

Properties: Diamond has a **very high melting point** due to its rigid structure. Since all the carbon atoms are bonded with other carbon atoms, no free electrons are present resulting in the structure being **non-conductive**. Diamond is a **non-conductor of electricity**.

2. Graphite: 1½ marks

Structure: Graphite is composed of flat **two-dimensional layers** of hexagonally arranged carbon atoms. In a layer, each C-atom is covalently bonded to **three other carbon atoms**. **Weak intermolecular**

bonds exist between each layer which allows the layers to slide over one another without breaking the bonds.

Properties: This arrangement makes graphite **soft and slippery**, making it ideal to be used as a lubricant. Graphite is a **good conductor of electricity** because it has free electrons (each carbon is bonded to only 3 others, leaving one electron free).

3. Buckyballs (Fullerenes / C-60): 1 mark

Structure: Buckyballs, also known as fullerenes, have a **football-like fused hollow ring structure** made up of **twenty hexagons and twelve pentagons**. Each of its **60 carbon atoms** is bonded to **3 other carbon atoms**.

Properties: They have a unique spherical cage-like structure. The name “buckyballs” comes from their resemblance to the geodesic domes designed by architect Buckminster Fuller.

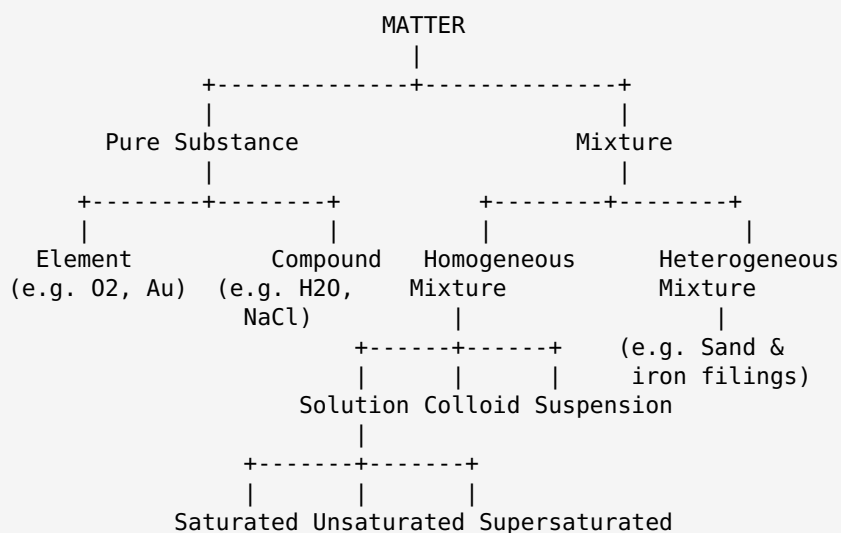
Q.3(iii): Differentiate between elements, compounds, and mixtures with examples. Explain the classification of matter with the help of a flowchart/diagram. 5 marks

Element: A pure substance made up of only **one type of atom**. It cannot be broken down into simpler substances by chemical means. Elements contain only one type of atom. Examples: Oxygen (O_2), Gold (Au), Carbon (C). 1 mark

Compound: A pure substance made up of **two or more elements chemically combined** in a fixed ratio. Compounds have different properties from the elements that make them and can only be separated into their elements through chemical reactions. Compounds consist of two or more elements chemically bonded in fixed proportions. Examples: Water (H_2O), Carbon dioxide (CO_2), Sodium chloride (NaCl). 1 mark

Mixture: A combination of two or more substances that are **physically mixed together** but not chemically combined. They do not have a fixed ratio. Mixtures consist of multiple substances that retain their individual properties and can be separated by physical processes. Examples: Air (mixture of nitrogen, oxygen, CO_2), Saltwater, Sand and iron filings. 1 mark

Classification of Matter (Flowchart): 2 marks



OR

Q.3(iii) OR: What is a solution? Describe the types of solutions (gaseous, liquid, solid) with at least two examples of each type. 5 marks

Solution: A solution is a **homogeneous mixture** of two or more substances in which one substance is dissolved in the other. Homogeneous means that no particles or parts of different substances can be seen. When one substance dissolves, the solution looks exactly the same. In solution, the particles are microscopic, less than 1 nm in diameter. A solution is very stable and the solute does not separate from

the solvent itself. **1 mark**

Solutions are found in three physical states depending upon the physical state of the solvent:

1. Gaseous Solutions: 1½ marks

In gaseous solutions, the solvent is a gas and the solute can be a gas, liquid, or solid.

- A **mixture of nitrogen and hydrogen** used in Haber's process (ammonia formation)
- A **mixture of ammonia and carbon dioxide** used for urea preparation
- **Fog, clouds, and mist** are examples where liquid water (solute) is dissolved in air (solvent)
- **Smoke** is a solution of carbon particles in gaseous air

2. Liquid Solutions: 1½ marks

In liquid solutions, the solvent is a liquid and the solute can be a gas, liquid, or solid.

- **Carbonated drinks** — solvent is liquid water and solute is gaseous carbon dioxide
- **Vinegar** (acetic acid in water) — liquid dissolved in liquid
- **Brine** (salt in water) and **sugar syrup** — solid dissolved in liquid

3. Solid Solutions: 1 mark

In solid solutions, the solvent is a solid and the solute can be a gas, liquid, or solid.

- **Hydrogen gas on nickel metal surface** — used in ghee industry (gas in solid)
- **Amalgam** — solution of any metal (solid) in liquid mercury
- **Alloys** (Brass, Bronze, Steel) — formed by mixing different metals (solid in solid)

Q.3(iv): Explain saturated, unsaturated, and supersaturated solutions with examples. Discuss the effect of temperature on solubility with reference to the solubility table. 5 marks

1. Saturated Solution: 1 mark

A solution containing **maximum amount of solute** at a given temperature is called a saturated solution. When solvent cannot dissolve any more solute at a given temperature, the solution is saturated. Further added solute remains undissolved and settles at the bottom. On the particle level, a saturated solution is one in which undissolved solute is in equilibrium with dissolved solute. At this stage, **dynamic equilibrium** is established — although dissolution and crystallization continue, the net amount of dissolved solute remains constant.

2. Unsaturated Solution: 1 mark

A solution which contains **lesser amount of solute** than that which is required to saturate it at a given temperature is called an unsaturated solution. Such solutions have the capacity to dissolve more solute to become a saturated solution.

3. Supersaturated Solution: 1 mark

When saturated solutions are heated, they develop further capacity to dissolve more solute. Such solutions contain **greater amount of solute** than is required to form a saturated solution and they become more concentrated. The solution that is more concentrated than a saturated solution is known as a **supersaturated solution**. Supersaturated solutions are **not stable**. An easy way to prepare one is to make a saturated solution at high temperature, then cool it to a temperature where excess solute crystallizes out. If you add a crystal of sodium thiosulphate to a supersaturated solution, crystallization will start.

Effect of Temperature on Solubility: 2 marks

Solubility is the maximum amount of solute which dissolves in a specified amount of solvent at a specific temperature. The solubility of solutes depends on temperature:

a) Increase in solubility with temperature: Most solid solutes become more soluble as temperature increases.

Solute	Solubility at 20°C (g/100g H ₂ O)	Solubility at 100°C (g/100g H ₂ O)
NaCl	36.5g	39.2g
KCl	34.7g	56g
NH ₄ Cl	37.5g	77g

b) Decrease in solubility with temperature: Some solutes become less soluble as temperature increases.

Solute	Solubility at 20°C	Solubility at 100°C
Ca(OH) ₂	0.173g	0.066g

Calcium hydroxide is an example whose solubility decreases with increasing temperature. It can be used to treat chemical burns and as an antacid.

OR

Q.3(iv) OR: Compare solutions, colloids, and suspensions in detail. Include particle size, visibility, filtration, Tyndall effect, and stability with examples. 5 marks

Solution: A solution is a **homogeneous mixture** of two or more components. Example: Sea water.

½ mark

Colloid: A colloid is a **heterogeneous mixture** in which the solute particles are larger than those in true solutions but not large enough to be seen by the naked eye. Particles range between 1 and 1000 nm in diameter and remain evenly distributed. They are also known as colloidal dispersions or false solutions. The Tyndall effect distinguishes colloids from solutions. Examples: Starch, albumin, soap solutions, blood, milk, ink, jelly, toothpaste. **½ mark**

Suspension: A heterogeneous mixture in which solid particles are spread throughout the liquid without dissolving. Particles are big enough (greater than 1000 nm) to be seen with naked eyes. Examples: Chalk in water (milky suspension), paints, milk of magnesia. **½ mark**

Detailed Comparison Table: 3½ marks

S.No	Property	Solution	Colloid	Suspension
1	Type	Homogeneous mixture	Heterogeneous mixture	Heterogeneous mixture
2	Particle size	Less than 1 nm. Not visible by naked eye	1 to 1000 nm (1-10 ³ nm). Not visible by naked eye	Greater than 10 ³ nm. Visible by naked eye
3	Filtration	Particles can pass through normal as well as ultra-filter paper	Particles can pass through normal filter paper but not through ultra-filter paper	Particles cannot pass through normal as well as ultra-filter paper
4	Tyndall effect	Cannot scatter the light (due to small size)	Can scatter the light (Tyndall effect)	Can scatter the light (Tyndall effect)
5	Stability	Does not separate	Does not separate	Separates or settles down when stationary
6	Examples	Sea water	Milk	Muddy water